

Small exact values and a dense-anchor bound for weak cube saturation

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Abstract

We study the weak saturation number of the three-dimensional cube Q_3 in the complete graph. By normalizing the first activated copy of Q_3 , we determine

$$\text{wsat}(K_8, Q_3) = 15, \quad \text{wsat}(K_9, Q_3) = 16, \quad \text{wsat}(K_{10}, Q_3) = 18.$$

The value at $n = 9$ corrects a reported value of 17 and disproves the associated conjecture $\text{wsat}(K_n, Q_3) = 2n - 1$. We also give an explicit 28-edge weakly saturated graph on 11 vertices containing a complete seven-vertex anchor. It yields

$$\text{wsat}(K_{N+4}, Q_3) \leq \text{wsat}(K_N, Q_3) + 7$$

and hence $\text{wsat}(K_n, Q_3) \leq \lfloor (7n + 2)/4 \rfloor$ for $n \geq 9$. Combined with a known combinatorial lower bound, this gives $11/7 \leq c_{Q_3} \leq 7/4$.

1 Weak saturation and the first cube

A spanning subgraph $H \subseteq K_n$ is weakly Q_3 -saturated if the missing edges can be added one at a time so that each added edge completes a new copy of Q_3 . Let $\text{wsat}(K_n, Q_3)$ be the minimum initial number of edges. We use the standard convention that the initial graph need not be Q_3 -free. Our explicit upper witnesses satisfy the stronger free condition, while the lower-bound searches do not assume it; therefore the values below are unchanged under the stricter convention.

Consider the first edge in any valid activation order. Before it is added, the initial graph already contains all eleven other edges of a copy of Q_3 . Since Q_3 is edge-transitive, this first $Q_3 - e$ witness may be fixed, up to isomorphism, on eight labelled vertices. Therefore an exact search at a proposed edge count only needs to enumerate choices outside one fixed 11-edge core.

Theorem 1.

$$\text{wsat}(K_8, Q_3) = 15, \quad \text{wsat}(K_9, Q_3) = 16, \quad \text{wsat}(K_{10}, Q_3) = 18. \quad (1)$$

Proof. For $n = 8$, exhaustive enumeration of the normalized 11-edge core plus zero through three additional edges rules out edge counts 11 through 14; a 15-edge witness closes to K_8 .

For $n = 9$, all normalized candidates through 15 edges were checked, and none closes. A 16-edge witness has a 20-step activation certificate closing it to K_9 .

For $n = 10$, any weakly saturated initial graph must be biconnected: adding an edge inside a copy of $Q_3 - e$ cannot merge distinct blocks because $Q_3 - e$ is biconnected, while the final graph is biconnected. Filtering by this necessary condition, all normalized candidates through 17 edges were exhausted. None closes, while an independently verified 18-edge witness closes to K_{10} .

The evidence layers are separate. The normalized candidate enumeration is implemented once by `exact_first_cube_enumeration.py`; the cube edge orbit and the biconnectivity condition are checked independently with NetworkX; and the K_9 , K_{10} , K_{11}/K_7 and K_{12} upper witnesses are stored as edge-by-edge certificates and replayed by a second NetworkX implementation. In particular, the 18-edge K_{10} witness has a 27-step certificate closing it to all 45 edges. This establishes (1). $\square$

A 2025 PCMI report [3] gave the correct value 15 at $n = 8$ but reported 17 at $n = 9$ and conjectured $2n - 1$ for all $n \geq 8$. The certified 16-edge graph corrects that value and disproves the conjecture. We do not identify the unreported program error.

2 Dense-anchor gluing

Lemma 2. *Let B be weakly Q_3 -saturated on b vertices, and let $A \subseteq V(B)$ have size $s \geq 2$. Put*

$$c(B, A) = |E(B)| - |E(B[A])|.$$

For every $N \geq \max\{8, s\}$,

$$\text{wsat}(K_{N+b-s}, Q_3) \leq \text{wsat}(K_N, Q_3) + c(B, A). \quad (2)$$

Proof. Identify A with s vertices of a weakly saturated graph on the old N vertices. Add the $b - s$ new vertices and precisely the edges of B having at least one new endpoint. First close the old graph to K_N . All edges inside A are then available, so the copy of B can be closed to K_b .

Choose distinct anchors $a_1, a_2 \in A$. For a new vertex p and an old vertex x still not adjacent to it, use p as one cube vertex and x, a_1, a_2 as its three neighbours. Four further old vertices complete a cube. Every required edge except px is already present, because the old vertices form a clique and pa_1, pa_2 are present after the K_b closure. Thus all cross edges may be added. The number of initially paid edges is $c(B, A)$. $\square$

3 A seven-edge four-vertex extension

Let $A = \{0, 1, \dots, 6\}$ span K_7 . Add four new vertices 7, 8, 9, 10, the path

$$7 - 8 - 9 - 10,$$

and the four spokes

$$3 - 7, \quad 4 - 8, \quad 0 - 9, \quad 5 - 10.$$

Together with the 21 anchor edges, this is a 28-edge graph $B \subset K_{11}$. An explicit edge-by-edge certificate verifies that it is weakly Q_3 -saturated and initially Q_3 -free. Here $b = 11$, $s = 7$, and $c(B, A) = 7$. Lemma 2 gives

$$\text{wsat}(K_{N+4}, Q_3) \leq \text{wsat}(K_N, Q_3) + 7 \quad (N \geq 8). \quad (3)$$

Using the exact seeds at $n = 9, 10$, a 19-edge seed at $n = 11$, and a certified 21-edge seed at $n = 12$, the four residue classes in (3) yield:

Theorem 3. *For every $n \geq 9$,*

$$\text{wsat}(K_n, Q_3) \leq \left\lfloor \frac{7n + 2}{4} \right\rfloor. \quad (4)$$

In particular,

$$c_{Q_3} := \lim_{n \rightarrow \infty} \frac{\text{wsat}(K_n, Q_3)}{n} \leq \frac{7}{4}.$$

Terekhov and Zhukovskii's Theorem 2.1 [1] uses the costs

$$(e_1, \dots, e_7) = (2, 4, 6, 7, 9, 10, 11)$$

for Q_3 . Its minimum additive closure has period seven: if $n - 8 = 7q + r$ with $0 \leq r < 7$, then

$$\text{wsat}(K_n, Q_3) \geq 11 + 11q + h_r, \quad (h_0, \dots, h_6) = (0, 2, 4, 6, 7, 9, 10).$$

The standard gluing/subadditivity argument recorded in their Section 2 gives the existence of c_{Q_3} . The seven-periodic formula follows by checking the seven residue classes in the min-plus recurrence for the costs e_i ; taking q parts of size seven and one remainder part attains the resulting lower bound. Hence the displayed specialization yields $c_{Q_3} \geq 11/7$, and

$$\frac{11}{7} \leq c_{Q_3} \leq \frac{7}{4}. \quad (5)$$

The lower bound in (5) is prior work, not a contribution of this note.

4 Reproducibility

The checks use Python 3.13.11 and NetworkX 3.6.1 from the project environment. From the project root, the exact lower-bound searches are

```
.venv\Scripts\python.exe loops\MATH-DIR-0053\verification\exact_first_cube_enumeration.py --n 8
↪ --maximum-edges 15
.venv\Scripts\python.exe loops\MATH-DIR-0053\verification\exact_first_cube_enumeration.py --n 9
↪ --maximum-edges 16
.venv\Scripts\python.exe loops\MATH-DIR-0053\verification\exact_first_cube_enumeration.py --n 10
↪ --maximum-edges 17 --structural-filter
```

Through the last excluded edge count, these enumerate respectively 697, 12,951 and 1,391,842 normalized candidates; in the $n = 10$ case the proved biconnectivity filter leaves 301,000 candidates for closure testing. The complete result files are `exact-first-cube-n8.json`, `exact-first-cube-n9.json`, and `exact-first-cube-n10.json` in `loops/MATH-DIR-0053/verification/results/`.

The K_9 , K_{10} , K_{11}/K_7 , and K_{12} upper certificates are independently replayed by

```
.venv\Scripts\python.exe loops\MATH-DIR-0053\verification\audit_activation_certificate.py
↪ <certificate.json>
```

using the four certificate files in the same results directory. The complete command list and the logical boundary between exhaustive search, structure checks, and certificate replay are recorded in `loops/MATH-DIR-0053/verification/README.md`.

5 Scope

The note does not determine $\text{wsat}(K_n, Q_3)$ for every n , nor the exact asymptotic constant. Overlapping-block gluing is also present in earlier weak-saturation work; the claimed increment is the concrete complete-anchor lemma, the certified K_{11}/K_7 block, the resulting $7/4$ upper bound, and the corrected exact values.

An attempted extension did not improve either side of (5). In particular, the standard single-interface reuse of the certified 21-edge K_{12} seed costs two edges per new vertex, and a uniform sparsity-matroid lower-bound scheme is capped at slope $11/7$. These are stopping results, not stronger bounds.

References

- [1] N. Terekhov and M. Zhukovskii, *Weak saturation in graphs: a combinatorial approach*, Journal of Combinatorial Theory, Series B **172** (2025), 146–167. [arXiv:2305.11043](https://arxiv.org/abs/2305.11043).
- [2] N. Terekhov and M. Zhukovskii, *Weak saturation rank: a failure of linear algebraic approach to weak saturation*, [arXiv:2405.17857](https://arxiv.org/abs/2405.17857).
- [3] T. Trakulthongchai, D. Xu and K. Zhu, *Weak saturation number of the 3-cube in the complete graph*, Park City Mathematics Institute, Summer 2025.
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